

Modeling of the hysteretic metal-insulator transition in a vanadium dioxide infrared detector

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Abstract. The vanadium dioxide (VO_2) thin film, usually employed as an infrared detector, exhibits hysteresis in its resistance-temperature characteristic. Considering a polycrystalline VO_2 thin film as a composite medium, containing semiconducting and metallic microcrystals, the well-known effective-medium approximation theory is employed to relate the volume fraction of the semiconducting microcrystals to the effective film resistance. A phenomenological model is first proposed for describing the hysteretic dependence of volume fraction on temperature. From this, a model for hysteresis in the resistance-temperature characteristic is then derived, and a procedure for estimating the model parameters is outlined. The model reproduces the more important hysteretic characteristics such as the major, minor, and nested loops, in good agreement with the experimental characteristics. © 2002 Society of Photo-Optical Instrumentation Engineers. [DOI: 10.1117/1.1501095]

Subject terms: vanadium dioxide; microbolometer; thermal hysteresis; sensor; infrared imaging; hysteresis model; minor loop; major loop; accommodation process; metal-insulator transition; composite medium; effective-medium approximation; percolation; phase transition.

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1 Introduction

Microbolometers, operating at room temperature and employing vanadium dioxide (VO_2) thin film as the heat-sensing material, have emerged as a promising solution for far-infrared imaging, providing a wide range of military, industrial, and commercial applications. The thermophysical properties of VO_2 thin films are characterized by a martensitic solid-state phase transformation at a temperature around 50°C . In this region, the material transforms from semiconducting at low temperatures to metallic at higher temperatures.

This phase transition is associated with significant variation of the structural, electrical, and optical material properties.¹ Recent fabrication techniques permit the deposition of vanadium dioxide films having a temperature coefficient of resistance [$\text{TCR} = (1/R)dR/dT$] of $3\% \text{ K}^{-1}$ in the semiconducting region² that increases to values as large as $200\% \text{ K}^{-1}$ in the hysteretic transition region.³ Despite this pronounced increase of the TCR, VO_2 -thin-film microbolometers are commonly operated outside the hysteretic region, in the semiconducting part of the resistance-temperature (R - T) curve.⁴

Experimental results, indicating a considerable increase in the sensitivity, have been recently reported for a VO_2 -thin-film microbolometer operating in the hysteretic region.³ The theoretical prediction of bolometric perfor-

mance in the transition region is quite complex, due to the hysteresis in the R - T characteristics. Thus, the analysis, design, and optimization of a VO_2 microbolometer operating in the hysteretic region is critically dependent on the availability of a hysteresis model that can mathematically describe both the major and minor loops.

The development of a hysteresis model, based on the Preisach's theory, for describing the behavior of a VO_2 thin film has been recently proposed.⁵ However, this model can describe only partially the R - T characteristic, which is quite asymmetrical. In the present work, a hysteresis model is proposed that describes more accurately the asymmetry of the hysteretic R - T characteristic of a VO_2 thin film. This model is expressed in the form of an algebraic equation and has six parameters, which can be readily determined from experimentally obtained major and minor R - T hysteresis loops.

2 Electrical Resistance of VO_2 Thin Films

The temperature-driven phase transition from a semiconducting to a metallic phase in a VO_2 thin film can be attributed to two types of coexisting crystallographic structures, or phases.¹ A typical VO_2 hysteresis curve, in the R - T plane, is shown in Fig. 1. During the first-order phase transition, more pronounced over the range of 35 to 65°C , the VO_2 microcrystals undergo a crystallographic transfor-

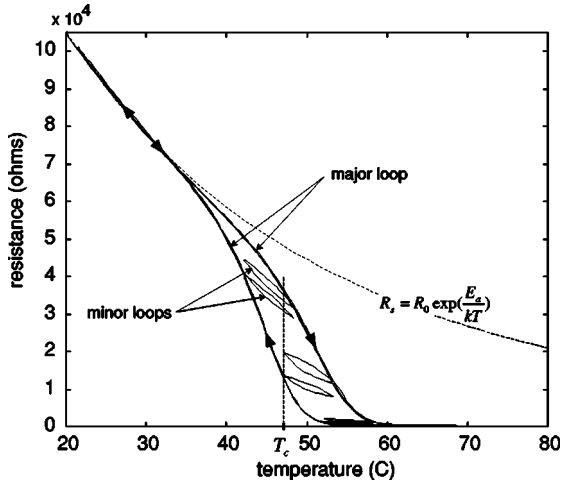


Fig. 1 Typical resistance-temperature characteristics of a VO₂ thin film, including minor loops. At low temperatures, the film resistance exhibits an exponential dependence on temperature. There is no bifurcation at low temperatures, and the behavior is fully reversible. At high temperatures, the film resistance is nearly constant (with a low value as for a metal), also describing reversible behavior. In the transition region, the resistance bifurcates into two different branches, revealing the irreversible characteristic of the film resistance in this region.

mation, accompanied by release or absorption of energy in the form of latent heat. For practical purposes, it is assumed that the phase transformation in the microcrystals occurs at temperatures between 20 and 80°C. For both semiconducting ($T < 20^\circ\text{C}$) and metallic ($T > 80^\circ\text{C}$) regions, the dependence of resistance on temperature is fully reversible, and release or absorption of energy is not observed.

2.1 Film Resistance in Semiconducting and Metallic Phases

In the semiconducting region of the R - T characteristic, the VO₂ microcrystals in the film behave as an intrinsic semiconductor, and the film resistance can be expressed as⁶

$$R_s = R_0 \exp\left(\frac{E_a}{kT}\right), \quad (1)$$

where T is the film temperature in kelvins, E_a is the activation energy, $k = 8.62 \times 10^{-5} \text{ eV K}^{-1}$ is the Boltzmann constant, and R_0 is the resistance for $T \rightarrow \infty$. The temperature coefficient of resistance then can be written as

$$\text{TCR} = \frac{1}{R_s} \frac{dR_s}{dT} = -\frac{E_a}{kT^2}. \quad (2)$$

For temperatures around 20°C, the activation energy E_a of VO₂ thin films is about 0.22 eV,² which is nearly the same as calculated from the measured $\text{TCR} = -3 \times 10^{-2} \text{ K}^{-1}$ (or $-3\% \text{ K}^{-1}$). For the samples employed in our experiments $R_0 = 17 \Omega$, as calculated from the measured value of the film resistance at $T = 20^\circ\text{C}$ employing Eq. (1). Thus, the film resistance in the semiconducting phase can be written as

$$R_s = 17 \exp\left(\frac{2553}{T}\right) \Omega. \quad (3)$$

In the metallic region ($T > 80^\circ\text{C}$), all VO₂ microcrystals behave as a metal and the film resistance R_m has nearly a constant value, almost independent of T . For the samples employed in our experiments, the film resistance R_m was measured at 80°C to be 140 Ω .

2.2 VO₂ Thin Film as a Composite Medium

A VO₂ thin film is a composite medium, formed by a large number of microcrystals, some of which are in the semiconducting phase and the rest in the metallic phase. The proportion between the volume fraction of semiconducting and metallic microcrystals in the film determines its electrical and optical properties.¹ When one of the constituent fractions in the film reaches a critical value, the percolation transition occurs.⁷ The percolation phenomenon can be explained as follows. When the film is heated up, many microscopic metallic islands appear on an insulating (semiconducting) substrate. With the increase in the number of metallic islands, conducting bridges between them appear, and some islands become electrically connected to each other. When the volume fraction of metallic particles reaches a certain value, the first conducting path is formed through the film. This is referred to as the percolation threshold. In percolation theory, the phase transition is described in terms of cluster formation;⁷ in the case of composite materials, it can also be described by employing the effective-medium approximation (EMA).⁸

In Ref. 1, the percolation phenomenon that occurs in VO₂ thin film and its consequences for the effective dielectric constant are explained in terms of the EMA. Employing the same approach, we propose to express the variation of electrical resistance of VO₂ thin films with temperature. The EMA theory treats each particle in a randomly inhomogeneous material as a particle embedded in a uniform medium whose effective conductivity σ_e represents the average conductivity of the compound. When employing the EMA, the resulting mathematical expressions to represent the effective conductivity σ_e depends on the basic assumptions concerning particle geometry and anisotropy.⁹ For the sake of simplicity, without loss of generality, the electrical resistance of a VO₂ thin film can be written as

$$R = gR_s + (1 - g)R_m, \quad (4)$$

where g is the volume fraction of the semiconducting microcrystals. In the case of VO₂ thin films, the resistance in the semiconducting phase is much higher than that in the metallic phase, and Eq. (4) may be simplified to

$$R = gR_s + R_m \quad (5)$$

$$= 17g \exp\left(\frac{2553}{T + 273}\right) + 140 \Omega, \quad (6)$$

where T is given in degrees Celsius.

It is interesting to point out that the application of the EMA results in a single expression (4) for the effective

resistance of this composite medium. However, the use of percolation theory for treating the phase transition in the composite medium would require two different scaling laws,⁷ one for volume fractions below the percolation threshold, and the other for volume fractions above it. This makes the mathematical formulation more complex.

3 Thermal Hysteresis in a VO₂ Thin Film

During the thermal cycling of a VO₂ thin film, the temperature at which the phase transformation of a microcrystal occurs is greatly affected by interfacial friction between neighboring grains,¹⁰ accounting for an irreversible component in the phase transition.

In the martensitic transformations¹¹ observed in VO₂ thin films, three energy components are involved: the chemical contribution, related to the structural or crystallographic change; the elastic mechanical energy; and the dissipative interfacial friction energy. The first two are considered reversible, while friction is responsible for the irreversible energy component. This irreversible component is the primary cause of the thermal hysteresis, which makes the dependence of film resistance on temperature multivalued.

3.1 Hysteresis Terminology

The study of hysteresis in VO₂ thin films requires the definition of some hysteresis terminology, as follows.

The curve in the R - T (or g - T) plane resulting from the increase of the film temperature from 20 to 80°C is termed the limiting *descending major curve*. The curve that results when the temperature is reduced from 80 to 20°C is termed the limiting *ascending major curve*. The combination of these two curves forms the limiting *major loop*, which is the principal hysteretic feature.

If, at any point along the descending or ascending major curve, dT/dt changes sign, the resulting trajectory in the R - T plane will approach respectively the ascending or descending major curve. This point is called *reversal point*, and the trajectory traversed is called a *first-order reversal curve*. Similarly, if along the first-order reversal curve another reversal point occurs, the resulting curve is called a *second-order reversal curve*, and so on. The closed trajectory that results from two consecutive higher-order curves is called a *minor loop*. The sequence of open trajectories formed by a sequence of higher-order curves is called a set of *nested minor loops*. When the temperature is continuously cycled back and forth between two identical end values, the resulting minor loop slowly drifts towards a final and stable equilibrium minor loop, after a few cycles. This phenomenon is normally observed for time-varying temperature excursions of small amplitude, and is called the *accommodation process*.

3.2 Dependence of Volume Fraction on Temperature

In the case of a VO₂ thin film, the typical major loop of the R - T characteristics is quite asymmetric. The presence of asymmetry creates mathematical difficulties associated with hysteresis modeling, and accurate modeling of a very

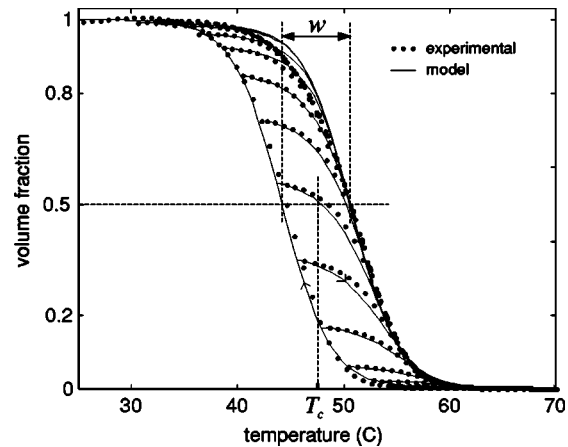


Fig. 2 Experimentally determined hysteresis characteristics for the volume-fraction evolution (solid circles) compared with those calculated from the model $F(T)$ (solid line).

asymmetric hysteresis curve is not an easy task. This may be verified in Ref. 5, where a considerable discrepancy between experimental and simulated data is observed for temperatures below 35°C.

As mentioned before, each grain, or microcrystal, within a polycrystalline VO₂ film has a specific temperature T at which its individual phase transition appears. The transition temperature of a microcrystal is determined by its size and volume, and is also affected by the interfacial friction between neighboring microcrystals. Present film fabrication techniques permit the deposition of vanadium dioxide films with selected grain size distribution. Consequently, associated with a certain fabrication method, there is a specific dependence of volume fraction g on temperature. In accordance with Eq. (6), and considering that R_s and R_m obey single-valued temperature dependence, the presence of hysteresis in the R - T characteristics implies hysteresis in the g - T characteristics. From Eq. (6), the experimental volume fraction $\tilde{g}$ can be determined as

$$\tilde{g} = \frac{1}{17} \frac{\tilde{R} - 140}{\exp\left(\frac{2553}{T + 273}\right)}, \quad (7)$$

where $\tilde{R}$ is the experimentally measured value (in ohms) of the film resistance at a temperature T (in degrees Celsius). The calculated volume-fraction $\tilde{g}$ - T characteristic is presented in Fig. 2.

We propose to obtain a functional description $g = F(T)$ for the hysteresis in the $\tilde{g}$ - T characteristics, and then employ $F(T)$ in (6) to obtain a hysteretic model for the film resistance $R(T)$. Thus, based on (6), the electrical resistance of a VO₂ thin film can be written as

$$R(T) = R_0 \exp\left(\frac{2553}{T + 273}\right) F(T) + R_m. \quad (8)$$

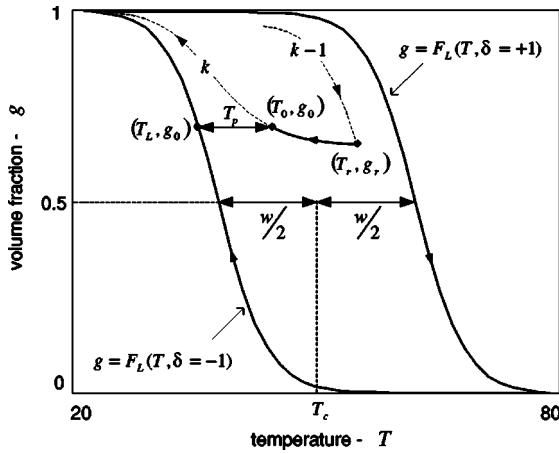


Fig. 3 Schematic representation of hysteresis in volume-fraction evolution with temperature. This geometric construction of a reversal curve illustrates the concept of proximity of a point on the reversal curve to the major loop. The major loop is formed by the combination of the limiting curves $F_L(T, \delta = +1)$ and $F_L(T, \delta = -1)$.

4 Modeling Hysteresis in Volume-Fraction Evolution

It can be observed in Fig. 2 that the descending and ascending $\tilde{g}$ - T major curves are separated by a certain width w . These curves saturate at $\tilde{g}=1$ for low temperatures and saturate at $\tilde{g}=0$ for high temperatures. In addition, all hysteresis trajectories are confined inside the major loop, which is the combination of the descending and ascending limiting (or major) curves. We propose the function $F_L(T)$ to describe the volume fraction g pertaining to the major descending and ascending limiting curves as

$$g = F_L(T) \triangleq \frac{1}{2} + \frac{1}{2} \tanh \left[\beta \left(\delta \frac{w}{2} + T_c - T \right) \right], \quad (9)$$

where w is the hysteresis width, β is related to dg/dT at T_c , and δ is a function defined as

$$\delta = \begin{cases} +1 & \text{for } dT/dt > 0, \\ -1 & \text{for } dT/dt < 0. \end{cases} \quad (10)$$

Here T_c is the critical temperature or percolation threshold, usually defined as the temperature for which $g=0.5$ (Fig. 3). In the case of hysteretic characteristics, there are two values for T_c , one for the descending and other for the ascending limiting curve. It is assumed that T_c is the average value of the temperatures for which $g=0.5$, at the center of the hysteresis curve. The combination of the curves $F_L(T, \delta = +1)$ and $F_L(T, \delta = -1)$ describes the major hysteresis loop in the g - T plane.

Reversal curves, minor loops, and nested minor loops are not described by Eq. (9). This equation has to be modified to represent the dependence of g on T for any trajectory inside the limiting major loop. To introduce this modification, it is particularly important to take into account the way in which the inner trajectories approach the major loop. This concept is illustrated in Fig. 3, where the change

in sign of dT/dt that occurs at the time instant t_r is shown. Hysteresis branch $k-1$ is reversed, and branch k starts at the reversal point (T_r, g_r) .

The *proximity temperature* T_p is introduced to express the nearness in distance of the current point (T_0, g_0) on branch k to the corresponding point (T_L, g_0) on the limiting curve $F_L(T, \delta)$, given as

$$T_p \triangleq T_L - T_0. \quad (11)$$

Using (9) we obtain the value of T_L corresponding to $g_0 = F_L(T_L, \delta)$ as

$$T_L = \delta \frac{w}{2} + T_c - \frac{1}{\beta} \operatorname{arctanh}(2g_0 - 1). \quad (12)$$

Thus, T_p at (T_0, g_0) is given from (11) as

$$T_p = \delta \frac{w}{2} + T_c - \frac{1}{\beta} \operatorname{arctanh}(2g_0 - 1) - T_0. \quad (13)$$

At the start of a new branch, at the reversal point (T_r, g_r) , T_p is termed as T_{pr} and can be written as

$$T_{pr} = \delta \frac{w}{2} + T_c - \frac{1}{\beta} \operatorname{arctanh}(2g_r - 1) - T_r. \quad (14)$$

To investigate the experimental behavior of T_p , the VO₂ film was thermally cycled and its electrical resistance measured for numerous minor and nested minor loops. These values of resistance were employed in (7) to determine the $\tilde{g}$ - T characteristics. The values of T_p were calculated employing (13) for the previously determined minor and nested minor loops in the g - T plane. It was observed from these experimental data that T_p exhibits almost the same functional dependence regardless of the reversal point (T_r, g_r) at which the reversal curve starts. Thus, to describe T_p for any hysteresis branch inside the major loop, the following functional dependence is proposed:

$$T_p \triangleq T_{pr} P(x), \quad (15)$$

where $x = (T_0 - T_r)/T_{pr}$, and $P(x)$ is an arbitrary monotonically decreasing function, with $P(0) = 1$, called here the *proximity function*.

Substituting (15) in (13) yields

$$T_{pr} P(x) = \delta \frac{w}{2} + T_c - \frac{1}{\beta} \operatorname{arctanh}(2g_0 - 1) - T_0. \quad (16)$$

Thus, the volume fraction corresponding to the current point (T_0, g_0) can be written from Eq. (16) as

$$g_0 = \frac{1}{2} + \frac{1}{2} \tanh \left(\beta \left[\delta \frac{w}{2} + T_c - \left[T_0 + T_{pr} P \left(\frac{T_0 - T_r}{T_{pr}} \right) \right] \right] \right), \quad (17)$$

and, for any arbitrary point (T, g) , the hysteretic volume fraction is expressed as

$$g = F(T) = \frac{1}{2} + \frac{1}{2} \tanh \left(\beta \left[\delta \frac{w}{2} + T_c - \left[T + T_{pr} P \left(\frac{T - T_r}{T_{pr}} \right) \right] \right] \right). \quad (18)$$

The values of δ , T_r , and T_{pr} undergo a change only at the reversal points, and remain unchanged until the next reversal occurs.

Initial attempts to use a low-order polynomial for $P(x)$ in Eq. (15) to fit the experimentally obtained dependence of T_p on T did not provide satisfactory results. Polynomials of higher order matched the measured data points, but exhibited highly oscillatory behavior between these points. This conflicts with the smooth hysteresis characteristics observed experimentally. On the other hand, the function $1 - \tanh kx$ provides a smooth characteristic, and fits quite well to the measured data near the reversal points, but fails for larger values of $T - T_r$. For the film employed in this work, we propose the following proximity function:

$$P(x) \triangleq \frac{1}{2} (1 - \sin \gamma x) [1 + \tanh(\pi^2 - 2\pi x)], \quad (19)$$

where γ is an arbitrary constant.

5 Results and Discussion

VO₂ thin films of approximately 1000-Å thickness, deposited on (100)-oriented semi-insulating Si substrates, were employed in this work. The films were fabricated by reactive Ar-ion magnetron sputtering. Samples were mounted on a controlled thermoelectric module. To obtain the model's parameters, the VO₂ thin film was subjected to a time-varying temperature excitation, composed of various monotonic segments. A digital temperature controller, implemented in a computer-supervised environment, was employed to obtain high-precision temperature ramping and measurement of the film resistance.¹²

From inspection of the calculated $\tilde{g}$ - T characteristics (Fig. 2), the values of w and T_c can be readily determined. It is not possible to determine β and γ from inspection of experimentally obtained curve.

The value of β can be obtained by fitting both $F_L(T, +1)$ and $F_L(T, -1)$ to the experimental major descending and ascending curves, respectively. Once the values of w , T_c , and β have been determined, the value of γ in $P(x)$ can be obtained by fitting Eq. (17) to a first-order descending $\tilde{g}$ - T curve. A first-order descending curve is generated by first increasing the temperature to 80°C, and then reducing the temperature monotonically until it reaches some value T such that the volume fraction $\tilde{g}$ lies on the ascending major curve. Subsequently, the temperature is again increased monotonically to 80°C. The choice of a descending curve, starting on the ascending major curve near its center ($T \approx 50^\circ\text{C}$), was found to be quite suitable for estimating γ . For the film employed in this work, the values obtained for the six parameters are: $w = 6.5$ K, $T_c = 47.6$ K, $\beta = 0.2$ K⁻¹, $\gamma = 0.85$, $R_0 = 17$ Ω and $R_m = 140$ Ω.

In Fig. 2, we present the experimental $\tilde{g}$ - T curves (solid circles) together with the volume-fraction g - T curves (solid

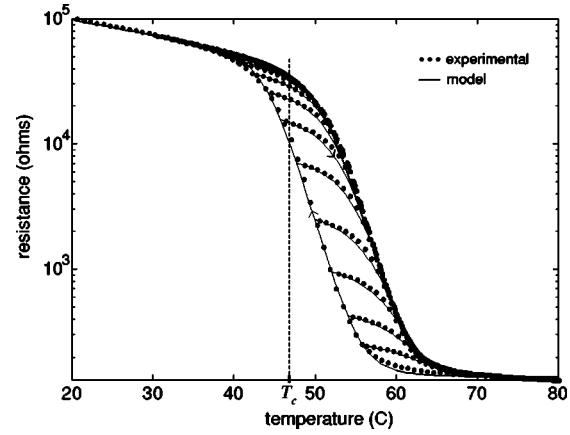


Fig. 4 Experimentally measured first-order descending R - T curves for the film resistance (solid circles) and those calculated from the model (8) (solid lines).

line) calculated from $F(T)$ employing (18). These two sets of curves were obtained for the same temperature excitation waveform. We observe in Fig. 2 that a discrepancy exists between the calculated and experimental curves in the region delimited by volume fractions ranging from 0.8 to 1.0 and temperatures ranging from 42 to 50°C. This discrepancy is due to a slight asymmetry of the experimental $\tilde{g}$ - T hysteresis curve in this region. Since the proposed model for the hysteretic volume fraction $g = F(T)$ is a hyperbolic-tangent function, which is symmetric in relation to its center ($T = T_c, g = 0.5$), this discrepancy depends on the level of asymmetry in the $\tilde{g}$ - T curve obtained experimentally.

To quantify the discrepancy between model and experimental data, we employ the *normalized root-mean-square error*, given as

$$e = \sqrt{\frac{\sum_{i=1}^N (x_i - \tilde{x}_i)^2}{\sum_{i=1}^N \tilde{x}_i^2}} \times 100\%, \quad (20)$$

where x_i represents data from the model, $\tilde{x}_i$ represents experimental data, and N is the number of data points. Using 216 data points, the mean error for all the first-order descending g - T curves (Fig. 2) was calculated to be 2.9%.

For evaluating microbolometric performance, we are particularly interested in the model for the R - T hysteresis loop, as given in Eq. (8). In Fig. 4, a set of experimentally obtained first-order descending $\tilde{R}$ - T curves (solid circles) is illustrated. It is from these curves that the $\tilde{g}$ - T curves, as depicted in Fig. 2, were calculated. The discrepancy observed between the measured $\tilde{R}$ - T curves and those (solid lines) obtained using the model given by Eq. (8) for $R(T)$, is visibly smaller than that observed in Fig. 2. In fact, the calculated mean error using Eq. (20) for all the first-order descending R - T curves, with a total of 539 data points, is 2.5%.

In Fig. 5, experimentally obtained first-order ascending R - T curves (solid circles), for a given temperature excitation waveform, as well as those (solid lines) calculated from Eq. (8) are presented. It is important to recall that the parameters of $F(T)$ in Eq. (8) were obtained from the ex-

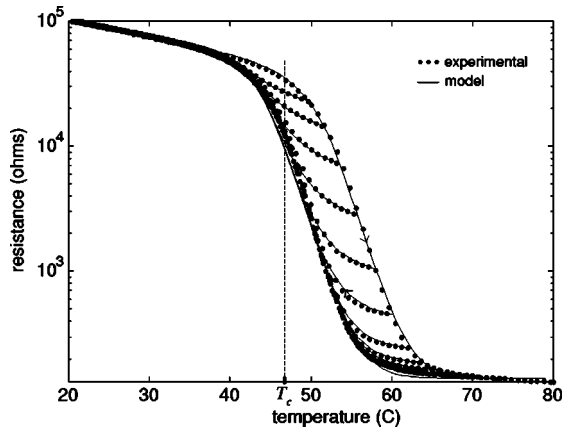


Fig. 5 Experimentally measured first-order ascending R - T curves for the film resistance (solid circles) and those calculated from the model (8) (solid lines).

perimental major loop and first-order descending $\tilde{g}$ - T curves. The calculated mean error for all the first-order ascending R - T curves, with a total of 694 data points, is 3.3%. The relatively good agreement observed between the experimental $\tilde{R}$ - T curves and those obtained from Eq. (8) validates the model.

The experimentally obtained higher-order reversal curves (solid circles), as well as those calculated from the model (solid lines), are shown in Fig. 6. A triangular waveform with progressively decreasing amplitude is employed for temperature excitation. From Fig. 6 we observe the ability of the model to represent these higher-order descending and ascending curves.

The ability of the model to describe minor loops can be observed in Fig. 7, where three pairs of minor loops are presented in three different regions. Each pair is composed of one minor loop emanating from the ascending major curve and the other emanating from the descending major curve. All minor loops were generated using the same temperature excursion, in different regions in the R - T plane. The minor loops pertaining to a particular pair have similar

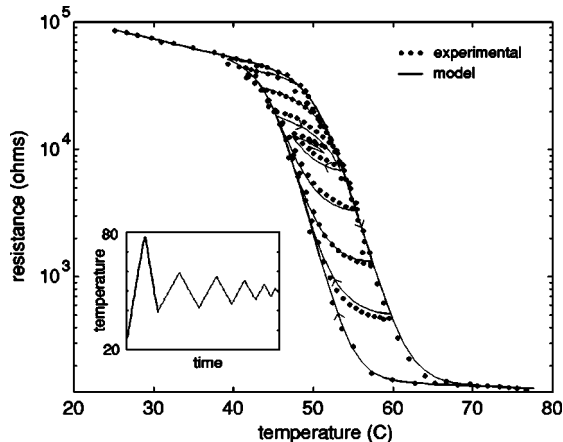


Fig. 6 Nested minor loops as calculated (solid lines) from the model (8) and those obtained experimentally (solid circles), using the temperature waveform shown in the inset. Temperature excitation period is 25 min.

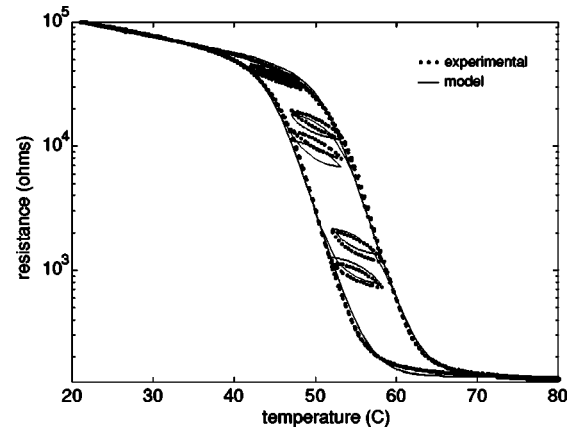


Fig. 7 Minor loops, emanating from the major descending and ascending curves, as calculated (solid lines) and as obtained experimentally (solid circles).

shapes, which vary from region to region. We observe a noticeable difference between experimental minor loops and those obtained from the model. This can be attributed to the high TCR in the transition region. The sample employed in this work exhibits a TCR of $60\% \text{ K}^{-1}$ in the transition region, which is equivalent to $dR/dT = 18 \text{ k}\Omega \text{ K}^{-1}$ for $R = 30 \text{ k}\Omega$, for example. With this dR/dT , even a small temperature measurement error of $\pm 0.1 \text{ K}$ can produce a difference as large as $\pm 1800 \Omega$ (6%) in the measured value of resistance as from that calculated from the model.

In Fig. 8(a), the experimentally observed accommodation process is depicted. The 1.0-K-wide minor loop drifts downwards, and after a few cycles it becomes stable. The accommodation phenomenon presented in Fig. 8(b) is as

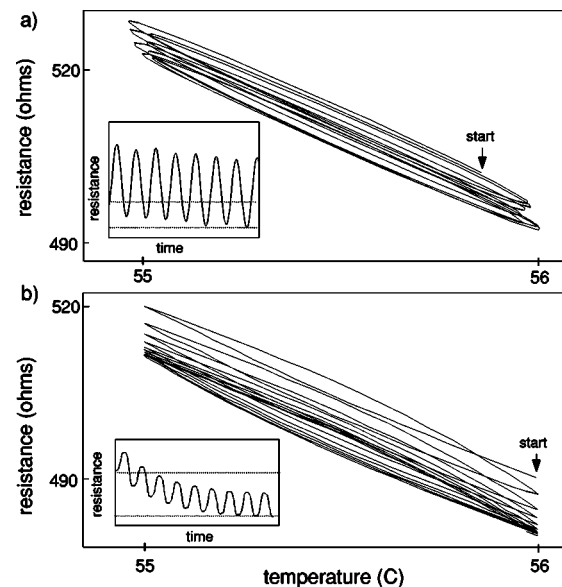


Fig. 8 (a) Experimentally observed accommodation process for a small-amplitude excursion in temperature, and (b) accommodation process calculated from the model (8) for the same temperature cycling. Temperature excitation period is 2 min.

calculated from the model, using the same excitation as in Fig. 8(a).

6 Conclusion

It has been established that it is possible to describe asymmetric hysteresis in the R - T characteristics in VO_2 -thin-film sensors by using EMA theory. For the calculation of the effective film resistance, the EMA requires the knowledge of the resistance and volume fraction of semi-conducting and metallic phases, which depend on temperature. From the measured hysteretic R - T characteristic, the hysteretic dependence of volume fraction g on temperature T was determined. A phenomenological model for the g - T characteristics was proposed. This model is expressed in the form of an algebraic equation that is computationally simple to implement. The parameters of this model were determined from the experimental data using a fairly simple and robust identification procedure. The model was validated against experimental data, with regard to major and minor loops, nested loops, and accommodation.

This paper deals only with the static features of the R - T characteristics of a VO_2 thin film. The effect of incident radiation is not addressed. We have also devised a model that, together with thermodynamic equilibrium equation, can be employed for microbolometer performance analysis even in the hysteretic transition region. This development is being carried out at present.

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References

1. H. Choi, J. Ahm, J. Jung, T. Noh, and D. Kim, *Phys. Rev. B* **54**(7), 4621–4628 (1996).
2. H. Jerominek, F. Picard, and D. Vincent, *Opt. Eng.* **32**(9), 2092–2099 (1993).
3. C. Reintsema, E. Grossman, and J. Koch, Eds., *Infrared Technology and Applications XXV*, *Proc. SPIE* **3698**, 190–200 (1999).
4. C. Chen, X. Yi, X. Zhao, and B. Xiong, in *25th International Conference on Infrared and Millimeter Waves*, pp. 211–212 (2000).
5. L. A. L. de Almeida, G. S. Deep, A. M. N. Lima, H. Neff, and R. C. S. Freire, *IEEE Trans. Instrum. Meas.* **50**(4), 1030–1035 (2001).
6. K. C. Liddiard, *Infrared Phys.* **26**(1), 43–49 (1986).
7. D. Stauffer and A. Aharony, *Introduction to Percolation Theory*, Taylor and Francis (1994).
8. D. Stroud, *Superlattices Microstruct.* **23**(3–4), 567–573 (1998).
9. T. W. Noh and P. H. Song, *Phys. Rev. B* **44**(11), 5459–5464 (1991).
10. P. Potapov, R. Gotthardt, and L. Bataillard, *Phys. Status Solidi A* **165**(2), 401–408 (1998).
11. K. Otsuka and C. M. Wayman, *Shape Memory Materials*, Cambridge University Press (1998).
12. A. M. N. Lima, G. S. Deep, L. A. L. de Almeida, H. Neff, and M. Fontana, in *Proc. 18th IEEE Instrumentation and Measurement Technology Conf.* **2**, 919–924 (2001).



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